

Subject: Programming with Java
ASSIGNMENT No :2

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BATCH:CSE B3

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TITLE :

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management

AIM:

To develop a Java application that demonstrates the use of default, parameterized, and copy constructors for object initialization and efficient data management.

PROGRAM CODE:

1.Create a Student class using default and parameterized constructors to initialize student name and roll number.

```

class Student {

    int id;
    String name;
    //Parameterized Constructor
    Student(int i, String n) {
        id = i;
        name = n;
    }
    //Copy Constructor
    Student(Student s) {
        id = s.id;
        name = s.name;
    }
    //default Constructor
    Student() {
        id = 194;
        name = "Suhani Garg";
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
    }

    public static void main(String[] args) {

        Student s1 = new Student();

        Student s2 = new Student(194, "Suhani");

        Student s3 = new Student(s2);

        System.out.println("Default Constructor:");
        s1.display();

        System.out.println();

        System.out.println("Parameterized Constructor:");
        s2.display();

        System.out.println();

        System.out.println("Copy Constructor:");
        s3.display();
    }
}

```

OUTPUT:

```

[suhanigarg670@Suhani-MacBook-Air desktop % javac Student.java
[suhanigarg670@Suhani-MacBook-Air desktop % java Student
Default Constructor:
ID: 194
Name: Suhani Garg

Parameterized Constructor:
ID: 194
Name: Suhani

Copy Constructor:
ID: 194
Name: Suhani
suhanigarg670@Suhani-MacBook-Air desktop %

```

QUESTIONS:

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```
class Mobile {  
    String brand;  
    String model;  
    int price;  
  
    Mobile() {  
        brand = "Samsung";  
        model = "Galaxy A35";  
        price = 30000;  
    }  
  
    Mobile(String b, String m, int p) {  
        brand = b;  
        model = m;  
        price = p;  
    }  
  
    Mobile(Mobile m) {  
        brand = m.brand;  
        model = m.model;  
        price = m.price;  
    }  
  
    void display() {  
        System.out.println("Brand: " + brand);  
        System.out.println("Model: " + model);  
        System.out.println("Price: " + price);  
    }  
  
    public static void main(String[] args) {  
        Mobile m1 = new Mobile();  
  
        Mobile m2 = new Mobile("Apple", "iPhone 16", 80000);  
  
        Mobile m3 = new Mobile(m2);  
  
        System.out.println("Default Constructor:");  
        m1.display();  
  
        System.out.println();  
  
        System.out.println("Parameterized Constructor:");  
        m2.display();  
  
        System.out.println();  
  
        System.out.println("Copy Constructor:");  
        m3.display();  
    }  
}
```

OUTPUT:

```
[suhanigarg670@Suhanis-MacBook-Air desktop % javac Mobile.java
[suhanigarg670@Suhanis-MacBook-Air desktop % java Mobile
Default Constructor:
Brand: Samsung
Model: Galaxy A35
Price: 30000

Parameterized Constructor:
Brand: Apple
Model: iPhone 16
Price: 80000

Copy Constructor:
Brand: Apple
Model: iPhone 16
Price: 80000
suhanigarg670@Suhanis-MacBook-Air desktop % █
```

CONCLUSION:

The experiment was successfully completed by implementing default, parameterized, and copy constructors in Java. The default constructor initialized objects with default values, the parameterized constructor initialized objects with user-defined values, and the copy constructor created duplicate objects by copying the data of an existing object. This experiment improved the understanding of object initialization, data management, and the role of constructors in object-oriented programming.